

Voice-Controlled Autonomous Navigation — DoggoBot

MAE/ECE 148 - Team 4



Summer Session II / 2026

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We Promised

A robot using camera-based computer vision and voice commands to execute a sequence of autonomous driving tasks

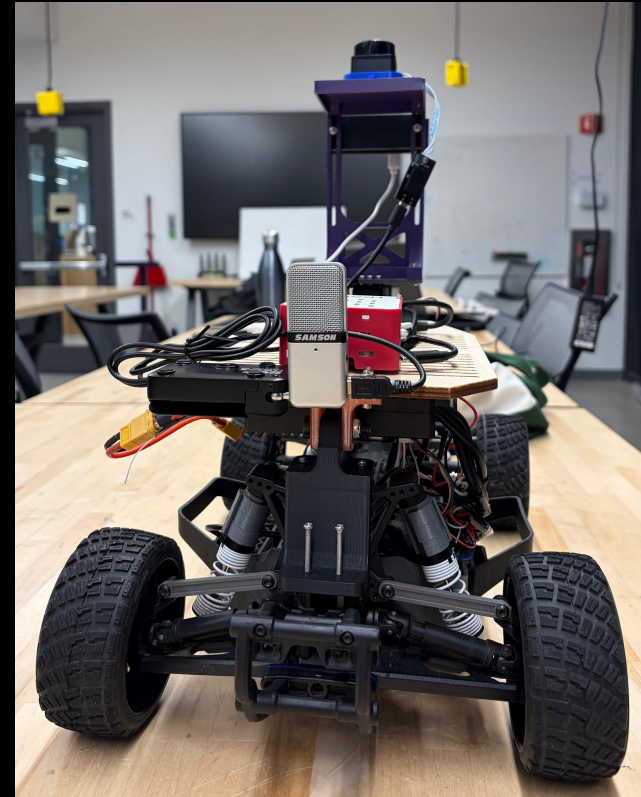
Must Haves

- Interpret simple voice commands
- Decipher and execute a string of commands
 - Using self-hosted LLM
- Real time human detection (YOLO AI)

Nice to Haves

- Remote controls via phone web app
- Follow commands
- 180 degree camera pivot
- Lidar collision protection

"Go forward for 3 seconds, stop, do a U-turn, then stop"



Primary Demonstration



Secondary Demonstration





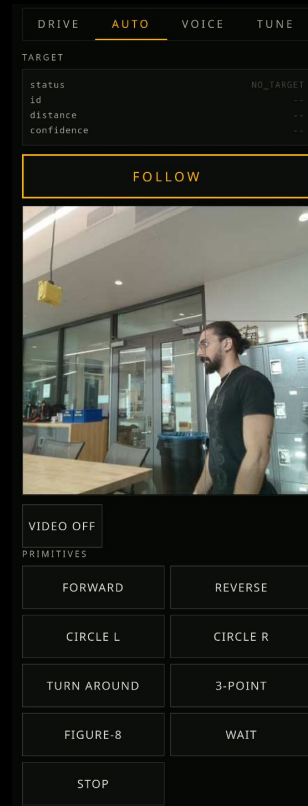
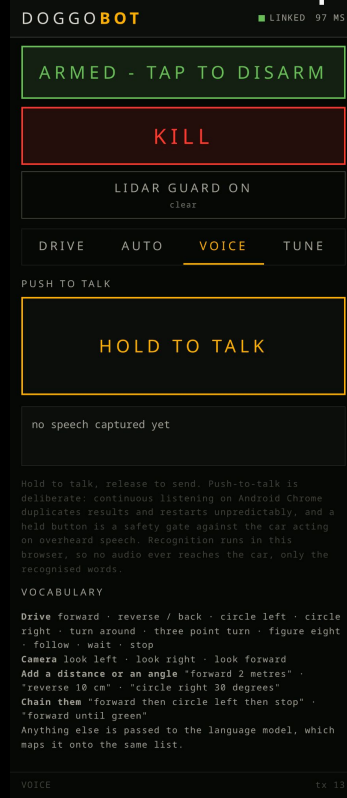
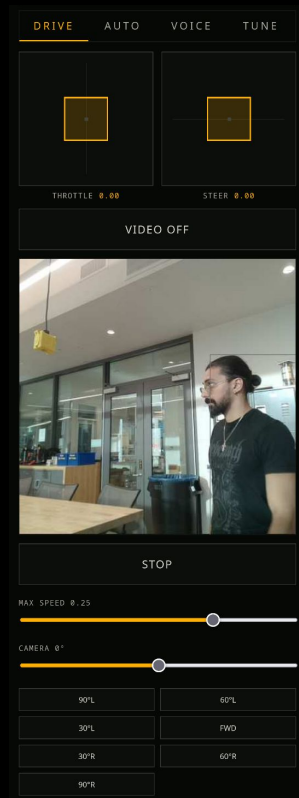
What Worked

- Execute simple and complex vocal commands
- Collision Protection
- Follow command (Human detection)
- Camera pivot, and it reacquires a lost target
- Phone web app remote controls with live video interface
- Distance and angle commands, measured not estimated

Roadblocks

- School wifi latency: campus wifi isolates clients. Moved to a private network.
- Onboard microphone noise (Cardioid mode + wake word)
- Cross calibration: every motion constant was a guess. Measured them: 0.635 m/s, 45°/s, 0.144 m coast

Phone Web App

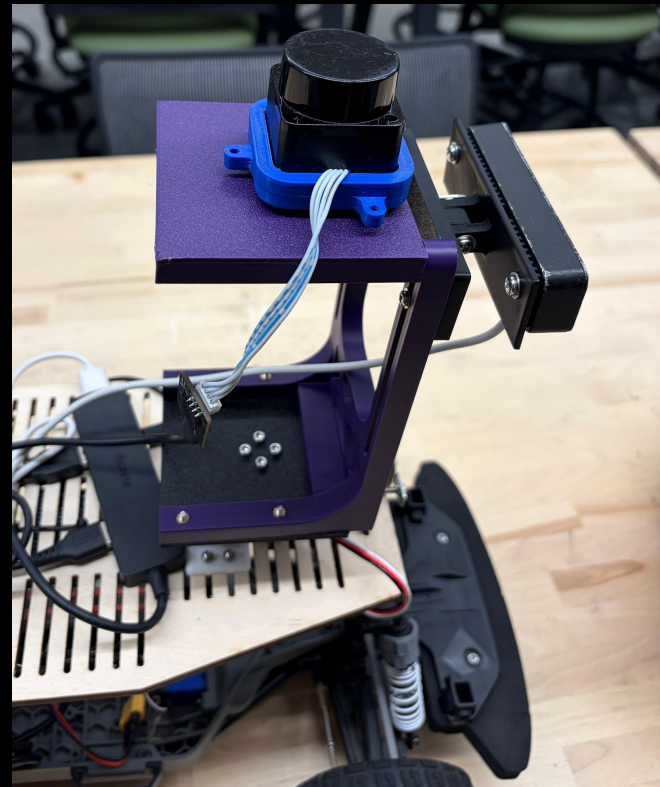


One HTML file, served by the robot itself. No install, works from any phone on the network.

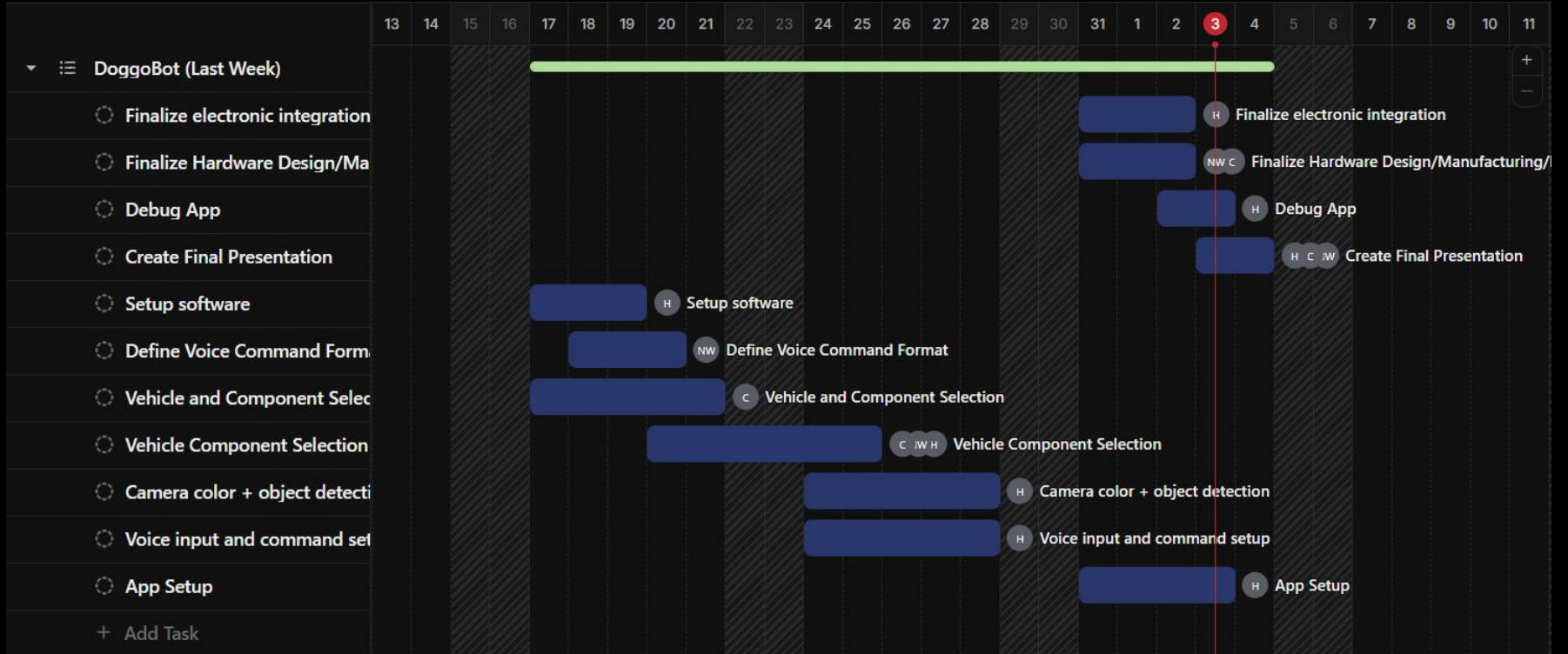


What we could've done better

- Better collision protection
 - Navigate *around* obstacles rather than only stopping for them
- More robust person detection
 - Detector runs at 12 fps; a faster loop would track a moving person better
 - Only the *person* class is loaded, so a coat rack can read as a person **at high confidence**
 - Cluttered rooms degrade it
- More robust following
 - The camera and steering cascade took significant tuning to stop it overshooting; more time would buy a better-behaved loop
- Wider detection
 - Facial recognition, so it knows its owner
 - Objects beyond people



Gantt Chart



Lessons Learned

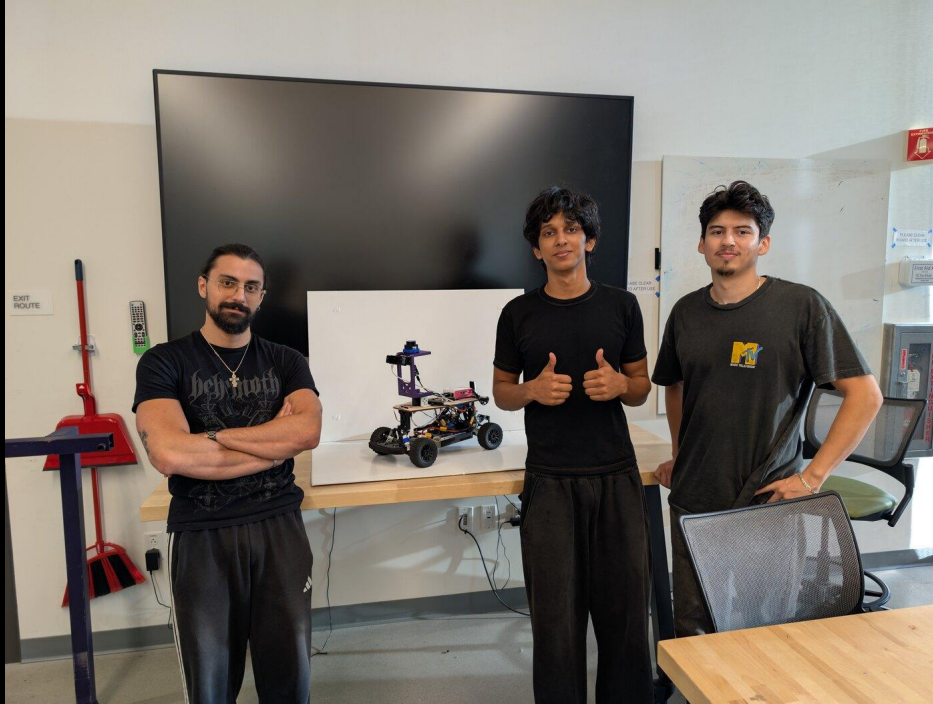
Measure don't estimate

Ensure enough wire and mechanical clearance

Capability has a compute budget

Confidence is not correctness

Thank you



Any Questions?

